

Notes: Bansal and Yaron (JF, 2004)

Risks for the Long Run: A Potential Resolution of Asset Pricing Puzzles

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 The asset-pricing facts the paper tries to match
- 2.2 Consumption, dividend, and asset-market data
- 2.3 Why the paper works with two cases
- 2.4 Key state variables
- 2.5 What “realized consumption volatility” means in the paper
- 2.6 Main calibration choices
- 2.7 Descriptive facts that matter most

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Recursive preferences and the Euler equation
- 3.2 Log-linear return representation
- 3.3 Case I: persistent expected growth only
- 3.4 Valuation ratios in Case I
- 3.5 Case I equity premium
- 3.6 Case II: adding stochastic volatility
- 3.7 Pricing-kernel innovation and time-varying premia
- 3.8 Volatility-feedback implication
- 3.9 Empirical predictive regressions used in the paper

4 Identification and credibility (what makes the model persuasive)

- 4.1 This is not an IV paper: credibility comes from internal discipline
- 4.2 Why low autocorrelations in consumption do not kill the long-run-risk idea

4.3	Why stochastic volatility is not just a free calibration device	
4.4	Why Epstein–Zin preferences are essential	
4.5	Why the i.i.d. benchmark fails	
4.6	What the paper still does not explain	
4.7	Conservative assumptions	
4.8	Relation to the literature	
5	Tables and figures	
5.1	Table I: annualized time-averaged growth rates	
5.2	Table II: asset-pricing implications, Case I	
5.3	Table III: properties of consumption volatility	
5.4	Table IV: asset-pricing implications, Case II	
5.5	Table V: decomposing the variance of the pricing kernel	
5.6	Table VI: predictability of returns, growth rates, and price-dividend ratios	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** Can a standard consumption-based asset-pricing model explain the equity premium, low risk-free rate, volatile valuations, time-varying return volatility, and long-horizon return predictability without relying on habits or market incompleteness?
- **Core diagnosis.** The paper argues that the standard i.i.d. consumption-growth benchmark throws away two economically important risks: news about *long-run expected growth* and news about *economic uncertainty*. Those risks are tiny in the time series, but asset prices can load heavily on them.
- **Main idea.** Model aggregate consumption and dividends with two ingredients: a small but highly persistent predictable growth component, and stochastic consumption volatility. Combine those dynamics with Epstein–Zin preferences so the intertemporal elasticity of substitution (IES) can exceed one even when risk aversion is also high.
- **Economic mechanism.** If the IES is above one, better long-run growth prospects raise valuation ratios. At the same time, higher economic uncertainty lowers valuation ratios and raises required risk premia. Because shocks to expected growth and volatility are persistent, even small news today changes long-horizon cash-flow and discount-rate beliefs enough to move prices a lot.
- **Why this is different from earlier models.** The paper does not rely on large short-run consumption shocks. Instead, it says that *long-run* risks matter: a very small persistent component in expected growth can generate large asset-price effects because it changes the full future path of consumption and dividends.
- **Main empirical support.** The paper shows that annual consumption and dividend data are consistent with a persistent growth component, and that realized consumption volatility is itself persistent and forecastable. Price-dividend ratios also predict future realized volatility with the correct sign.
- **Main quantitative result.** With risk aversion around 10 and IES around 1.5, the model can jointly match a large equity premium, a low risk-free rate, realistic market-return volatility, realistic risk-free-rate volatility, persistent price-dividend ratios, and long-horizon return predictability.
- **Key decomposition result.** In the preferred specification, about half of price-dividend-ratio variation comes from expected-growth news and the rest from time-varying economic uncertainty. So the paper is not saying valuations are *only* discount-rate variation.
- **Key asymmetry in the economics.** Good long-run growth news raises prices, while volatility news lowers prices. This generates a volatility-feedback effect: return news and future-volatility news are negatively correlated.
- **Main credibility point.** The paper compares the long-run-risk model against an i.i.d. benchmark and shows that the i.i.d. benchmark fails badly. The persistent-growth and stochastic-volatility ingredients are not decorative. They are exactly what produces the asset-pricing improvements.
- **Main limitation.** The paper does *not* generate the negative relation between expected returns and market volatility discussed in part of the literature. The authors are explicit that this

remains outside the model.

- **Takeaway.** The paper’s message is that asset markets care a lot about rare but persistent news in macro fundamentals. Once expected growth and uncertainty are allowed to move slowly over time, many “puzzles” become much less puzzling.

2 Data and measurement (key objects)

2.1 The asset-pricing facts the paper tries to match

The paper starts from a familiar list of difficult moments for standard models:

- an equity premium around 6% per year,
- a low average risk-free rate,
- market-return volatility around 19% per year,
- persistent variation in the price-dividend ratio,
- long-horizon return predictability from the dividend yield,
- and time-varying return volatility.

A further target is the negative relation between consumption volatility and valuation ratios: when macroeconomic uncertainty rises, prices should fall.

2.2 Consumption, dividend, and asset-market data

The paper calibrates the model at the **monthly** decision frequency but evaluates it against **annual** U.S. data from **1929 to 1998**. The main empirical series are:

- real per-capita consumption growth of nondurables plus services,
- dividend growth from the CRSP value-weighted market return,
- the market return,
- the risk-free rate,
- and the aggregate price-dividend ratio.

The authors time-aggregate the monthly model to annual frequency before comparing moments. They also use Monte Carlo simulations of samples with the same length as the data so that finite-sample distortions in autocorrelations and variance ratios are treated symmetrically for the model and the data.

2.3 Why the paper works with two cases

The quantitative section is organized around two nested versions of the model:

- **Case I:** persistent expected growth only,

- **Case II:** persistent expected growth plus stochastic volatility.

This is useful because it lets the paper isolate what the volatility channel adds on top of the long-run-growth channel.

2.4 Key state variables

The paper’s macro state variables are:

- x_t : the small persistent component in expected consumption growth,
- σ_t^2 : conditional variance of consumption growth, interpreted as macroeconomic uncertainty.

Consumption growth is only weakly predictable in any single month, but because x_t is highly persistent, its asset-pricing effect can be large. Dividends are allowed to respond more strongly than consumption to long-run growth news, which makes equities more levered claims on the state variables.

2.5 What “realized consumption volatility” means in the paper

The paper measures realized volatility from annual consumption-growth residuals. Specifically, it looks at the absolute value of the innovation in annual consumption growth after projecting growth on its own lags. If volatility were i.i.d., the variance ratios of this realized-volatility measure would be flat across horizons. The data instead show a rising pattern, which the authors interpret as evidence of persistent volatility.

2.6 Main calibration choices

The preferred parameter values are deliberately moderate rather than extreme:

- monthly discount factor $\delta = 0.998$,
- risk aversion $\gamma = 10$,
- IES $\psi = 1.5$,
- persistence of expected growth $\rho = 0.979$,
- monthly consumption-shock scale $\sigma = 0.0078$,
- leverage of dividend growth with respect to expected growth $\phi = 3$,
- additional dividend-shock loading $\varphi_d = 4.5$,
- volatility persistence $\nu_1 = 0.987$,
- volatility-shock scale $\sigma_w = 0.23 \times 10^{-5}$.

The point is not that every parameter is estimated structurally. The point is that this configuration jointly respects the time-series data and produces plausible asset-pricing moments.

2.7 Descriptive facts that matter most

The paper emphasizes four empirical facts.

- **Consumption growth is hard to distinguish from i.i.d. in finite samples**, even if a small persistent component is present.
- **Dividend growth is much more volatile than consumption growth**, which is why equities amplify long-run risks.
- **Realized consumption volatility is persistent**, as seen in variance-ratio evidence.
- **Price-dividend ratios forecast future realized volatility with a negative sign**, meaning high valuations are associated with lower future uncertainty.

These moments are exactly the ones the model is built to explain.

3 Models and empirical specifications (all equations + interpretation)

3.1 Recursive preferences and the Euler equation

The representative agent has Epstein–Zin recursive preferences. The asset-pricing restriction is

$$E_t \left[\delta^\theta G_{t+1}^{-\theta/\psi} R_{a,t+1}^{-(1-\theta)} R_{i,t+1} \right] = 1, \quad \theta = \frac{1-\gamma}{1-1/\psi}. \quad (1)$$

Here G_{t+1} is gross consumption growth, $R_{a,t+1}$ is the return on the consumption claim, $R_{i,t+1}$ is any asset return, γ is risk aversion, and ψ is the IES.

Interpretation. Epstein–Zin preferences matter because they separate risk aversion from the willingness to substitute consumption intertemporally. That separation is what lets the model have both high risk premia and a low, not-too-volatile risk-free rate.

3.2 Log-linear return representation

Following Campbell–Shiller log-linearization, the return on the aggregate consumption claim is approximated as

$$r_{a,t+1} \approx \kappa_0 + \kappa_1 z_{t+1} - z_t + g_{t+1}, \quad (2)$$

where $z_t = \log(P_t/C_t)$ is the log price-consumption ratio and g_{t+1} is log consumption growth.

The log intertemporal marginal rate of substitution is

$$m_{t+1} = \theta \log \delta - \frac{\theta}{\psi} g_{t+1} + (\theta - 1) r_{a,t+1}. \quad (3)$$

Interpretation. Any asset earns a premium only to the extent that its return covaries negatively with m_{t+1} . The problem therefore reduces to asking what shocks move consumption growth and valuation ratios, and how persistent those shocks are.

3.3 Case I: persistent expected growth only

The first version of the model assumes a small persistent component in expected growth:

$$\begin{aligned}x_{t+1} &= \rho x_t + \varphi_e \sigma e_{t+1}, \\g_{t+1} &= \mu + x_t + \sigma \eta_{t+1}, \\g_{d,t+1} &= \mu_d + \phi x_t + \varphi_d \sigma u_{t+1},\end{aligned}\tag{4}$$

with $e_{t+1}, u_{t+1}, \eta_{t+1} \sim \text{i.i.d. } N(0, 1)$.

Here x_t is the persistent long-run growth state, g_{t+1} is consumption growth, and $g_{d,t+1}$ is dividend growth.

Interpretation.

- x_t is tiny but persistent, so a shock to x_t changes expected growth far into the future.
- Dividends respond more strongly than consumption when $\phi > 1$, so equity prices react more than the price of the consumption claim.
- The model is deliberately conservative about short-run shocks: the main action is in persistence, not in huge innovation variances.

3.4 Valuation ratios in Case I

The solution takes the form

$$z_t = A_0 + A_1 x_t, \quad z_{m,t} = A_{0,m} + A_{1,m} x_t,\tag{5}$$

where $z_{m,t} = \log(P_t/D_t)$ is the log price-dividend ratio. The key coefficients are

$$A_1 = \frac{1 - 1/\psi}{1 - \kappa_1 \rho}, \quad A_{1,m} = \frac{\phi - 1/\psi}{1 - \kappa_{1,m} \rho}.\tag{6}$$

Interpretation.

- If $\psi > 1$, then $A_1 > 0$: better expected long-run growth raises the wealth-consumption ratio.
- If $\phi > 1$, then $A_{1,m} > A_1$: dividend claims are more sensitive than consumption claims to long-run growth news.
- This is the core long-run-risk amplification mechanism.

3.5 Case I equity premium

In the persistent-growth model, the market equity premium can be written in reduced form as

$$E_t(r_{m,t+1} - r_{f,t}) = \beta_{m,e} \lambda_{m,e} \sigma^2 - \frac{1}{2} \text{Var}_t(r_{m,t+1}),\tag{7}$$

where $\beta_{m,e}$ is the exposure of equity returns to long-run growth news and $\lambda_{m,e}$ is the price of that risk.

Interpretation. The premium is high when expected-growth shocks both move equity prices a lot and are highly priced by the pricing kernel. Persistence raises both objects.

3.6 Case II: adding stochastic volatility

The full model augments the system with time-varying macro uncertainty:

$$\begin{aligned}x_{t+1} &= \rho x_t + \varphi_e \sigma_t e_{t+1}, \\g_{t+1} &= \mu + x_t + \sigma_t \eta_{t+1}, \\g_{d,t+1} &= \mu_d + \phi x_t + \varphi_d \sigma_t u_{t+1}, \\\sigma_{t+1}^2 &= \bar{\sigma}^2 + \nu_1(\sigma_t^2 - \bar{\sigma}^2) + \sigma_w w_{t+1}.\end{aligned}\tag{8}$$

The pricing functions become

$$z_t = A_0 + A_1 x_t + A_2 \sigma_t^2, \quad z_{m,t} = A_{0,m} + A_{1,m} x_t + A_{2,m} \sigma_t^2.\tag{9}$$

Interpretation.

- σ_t^2 is the paper's measure of economic uncertainty.
- When $\gamma > 1$ and $\psi > 1$, the volatility coefficients are negative: higher uncertainty lowers valuation ratios.
- Persistence in σ_t^2 matters for the same reason persistence in x_t matters: it makes today's uncertainty shock a long-run object.

3.7 Pricing-kernel innovation and time-varying premia

With stochastic volatility, the innovation in the pricing kernel loads on three shocks:

$$m_{t+1} - E_t(m_{t+1}) = \lambda_{m,\eta} \sigma_t \eta_{t+1} - \lambda_{m,e} \sigma_t e_{t+1} - \lambda_{m,w} \sigma_w w_{t+1}.\tag{10}$$

The market equity premium becomes

$$E_t(r_{m,t+1} - r_{f,t}) = \beta_{m,e} \lambda_{m,e} \sigma_t^2 + \beta_{m,w} \lambda_{m,w} \sigma_w^2 - \frac{1}{2} \text{Var}_t(r_{m,t+1}).\tag{11}$$

Interpretation.

- Expected-growth risk still matters.
- Now volatility risk matters too, so the equity premium and Sharpe ratio become time-varying.
- During high-uncertainty periods, required returns rise and prices fall.

3.8 Volatility-feedback implication

A central implication is that return news and future-volatility news are negatively related. The paper shows the model-implied correlation between market-return innovations and return-volatility

innovations is about -0.32 .

Interpretation. This is the classic “bad news raises required returns” channel. It helps the model generate realistic variation in valuations rather than just in expected cash flows.

3.9 Empirical predictive regressions used in the paper

The paper evaluates the model with three time-series exercises.

First, return predictability:

$$\sum_{s=1}^j r_{t+s}^e = a(j) + B(j) \log(P_t/D_t) + v_{t+j}. \quad (12)$$

Because the regressor is the log *price-dividend* ratio, the coefficients are negative when a higher dividend yield predicts higher future returns.

Second, consumption-growth predictability:

$$\sum_{s=1}^j g_{t+s} = a(j) + B(j) \log(P_t/D_t) + v_{t+j}. \quad (13)$$

Third, two regressions that connect valuations and realized volatility:

$$|\hat{\varepsilon}_{g,t+j}| = a(j) + B(j)(p_t - d_t) + v_{t+j}. \quad (14)$$

$$(p_{t+j} - d_{t+j}) = a(j) + B(j)|\hat{\varepsilon}_{g,t}| + v_{t+j}. \quad (15)$$

Interpretation.

- Equation (12) asks whether valuation ratios forecast long-horizon excess returns.
- Equation (13) asks whether they forecast long-horizon consumption growth.
- Equations (14) and (15) ask whether uncertainty is persistent and negatively related to valuations.

4 Identification and credibility (what makes the model persuasive)

4.1 This is not an IV paper: credibility comes from internal discipline

The paper has no external instrument. Its credibility comes from a different structure:

- the model is transparent and parsimonious,
- the economic channels are isolated one by one (Case I versus Case II),
- the same parameterization is required to match both macro moments and asset-pricing moments,

- and the model is checked against predictive regressions rather than just unconditional averages.

So the paper’s standard is not “causal identification” in the policy-evaluation sense. The standard is whether a disciplined equilibrium model can simultaneously organize a broad set of facts.

4.2 Why low autocorrelations in consumption do not kill the long-run-risk idea

A natural objection is that if expected growth is persistent, realized consumption growth should look highly persistent too. The paper’s answer is that the predictable component is *small* and therefore difficult to detect in finite samples. Shephard and Harvey’s point is that a small persistent component can look observationally close to i.i.d. growth in short samples.

Why this matters. The i.i.d. model and the long-run-risk model can look similar in raw macro time-series tests, yet have completely different asset-pricing implications. This is the central logic behind Table I.

4.3 Why stochastic volatility is not just a free calibration device

The paper does not simply assume persistent volatility and stop there. It provides direct evidence that realized consumption volatility is persistent and that the price-dividend ratio predicts future realized volatility with a negative sign.

Why this matters. The uncertainty state is tied to data moments outside the core asset-pricing targets. This makes the volatility channel much more convincing than a purely latent device chosen only to fit returns.

4.4 Why Epstein–Zin preferences are essential

The paper’s preferred calibration has both $\gamma > 1$ and $\psi > 1$. That combination is crucial.

- If $\psi < 1$, higher expected growth can lower valuation ratios because the wealth effect dominates the substitution effect.
- If growth is i.i.d., the model loses the long-run channel that produces large valuation sensitivity and a realistic premium.
- With standard expected utility and no long-run risks, the paper inherits the classic problems stressed by Mehra–Prescott and Weil.

Bottom line. The paper is not saying “Epstein–Zin alone solves the puzzle.” It is saying that Epstein–Zin *combined with long-run risks* changes the economics decisively.

4.5 Why the i.i.d. benchmark fails

Panel C of Table II is especially important. When consumption and dividend growth are made i.i.d., the model produces a *negative* equity premium of about -0.74% in the baseline calibration and zero variation in the price-dividend ratio. So the benchmark model fails exactly where the long-run-risk model succeeds.

Interpretation. This is the sharpest model-comparison exercise in the paper. It shows that the persistent-growth state is not a small technical embellishment; it is the main source of the improvement.

4.6 What the paper still does not explain

The paper openly acknowledges three limitations.

- It does not generate the negative relation between expected returns and market volatility discussed by some empirical papers.
- Predictive-regression evidence is noisy and sample-sensitive, so some successes should be interpreted cautiously.
- The model works with cash dividends, which may be a poor measure of total payouts and can distort growth-predictability tests.

This candor is important: the paper solves many puzzles, but not every asset-pricing fact.

4.7 Conservative assumptions

Several assumptions actually work *against* the model rather than in its favor.

- Innovations to consumption growth, dividends, and expected growth are assumed independent in the baseline.
- The correlation between dividends and consumption is calibrated conservatively.
- The paper uses cash dividends rather than broader payout measures.

The authors show that allowing modest correlations between shocks would typically make the equity premium larger, not smaller.

4.8 Relation to the literature

The paper's place in the literature is best understood relative to four benchmarks.

- **Mehra–Prescott / Weil:** the standard consumption model fails because i.i.d. growth contains too little priced risk.
- **Barsky–DeLong:** persistence in expected growth matters for valuations, but Bansal–Yaron bring this into an equilibrium asset-pricing model.
- **Kandel–Stambaugh:** allowing predictability and volatility helps, but the persistence and IES configuration in Bansal–Yaron make the quantitative fit much stronger.
- **Campbell–Cochrane:** both papers address puzzles, but Bansal–Yaron do it through long-run macro risks rather than external habit.

5 Tables and figures

This paper is almost entirely **table-driven**. The empirical and quantitative evidence is concentrated in Tables I–VI rather than in a separate figure set.

5.1 Table I: annualized time-averaged growth rates

Read:

- The model is calibrated so annual consumption growth has volatility close to the data: the data value is about 2.93%, and the model population value is about 2.88%.
- First-order autocorrelation in consumption growth is around 0.49 in the data and 0.53 in the model population.
- The model also reproduces the broad pattern of variance ratios and the much higher volatility of dividend growth.

Economic message. The long-run-growth state is hard to reject from macro data alone. The model looks broadly consistent with annual consumption and dividend moments even though its asset-pricing implications are very different from the i.i.d. benchmark.

Table I
Annualized Time-Averaged Growth Rates

The model parameters are based on the process given in equation (4). The parameters are $\mu = \mu_d = 0.0015$, $\rho = 0.979$, $\sigma = 0.0078$, $\phi = 3$, $\psi_e = 0.044$, and $\psi_d = 4.5$. The statistics for the data are based on annual observations from 1929 to 1998. Consumption is real nondurables and services (BEA); dividends are from the CRSP value-weighted return. The expression $AC(j)$ is the j^{th} autocorrelation, $VR(j)$ is the j^{th} variance ratio, and $corr$ denotes the correlation. Standard errors are Newey and West (1987) corrected using 10 lags. The statistics for the model are based on 1,000 simulations each with 840 monthly observations that are time-aggregated to an annual frequency. The *mean* displays the mean across the simulations. The 95% and 5% columns display the estimated percentiles of the simulated distribution. The *p-val* column denotes the number of times in the simulation the parameter of interest was larger than the corresponding estimate in the data. The *Pop* column refers to population value.

Variable	Data		Model				
	Estimate	SE	Mean	95%	5%	<i>p-Val</i>	<i>Pop</i>
$\sigma(g)$	2.93	(0.69)	2.72	3.80	2.01	0.37	2.88
$AC(1)$	0.49	(0.14)	0.48	0.65	0.21	0.53	0.53
$AC(2)$	0.15	(0.22)	0.23	0.50	-0.17	0.70	0.27
$AC(5)$	-0.08	(0.10)	0.13	0.46	-0.13	0.93	0.09
$AC(10)$	0.05	(0.09)	0.01	0.32	-0.24	0.80	0.01
$VR(2)$	1.61	(0.34)	1.47	1.69	1.22	0.17	1.53
$VR(5)$	2.01	(1.23)	2.26	3.78	0.79	0.63	2.36
$VR(10)$	1.57	(2.07)	3.00	6.51	0.76	0.77	2.96
$\sigma(g_d)$	11.49	(1.98)	10.96	15.47	7.79	0.43	11.27
$AC(1)$	0.21	(0.13)	0.33	0.57	0.09	0.53	0.39
$corr(g, g_d)$	0.55	(0.34)	0.31	0.60	-0.03	0.07	0.35

Table II
Asset Pricing Implications—Case I

This table provides information regarding the model without fluctuating economic uncertainty (i.e., Case I, where $\sigma_w = 0$). All entries are based on $\delta = 0.998$. In Panel A the parameter configuration follows that in Table I, that is, $\mu = \mu_d = 0.0015$, $\rho = 0.979$, $\sigma = 0.0078$, $\phi = 3$, $\psi_e = 0.044$, and $\psi_d = 4.5$. Panels B and C describe the changes in the relevant parameters. The expressions $E(R_m - R_f)$ and $E(R_f)$ are, respectively, the annualized equity premium and mean risk-free rate. The expressions $\sigma(R_m)$, $\sigma(R_f)$, and $\sigma(p - d)$ are the annualized volatilities of the market return, risk-free rate, and the log price-dividend, respectively.

γ	ψ	$E(R_m - R_f)$	$E(R_f)$	$\sigma(R_m)$	$\sigma(R_f)$	$\sigma(p - d)$
Panel A: $\phi = 3.0$, $\rho = 0.979$						
7.5	0.5	0.55	4.80	13.11	1.17	0.07
7.5	1.5	2.71	1.61	16.21	0.39	0.16
10.0	0.5	1.19	4.89	13.11	1.17	0.07
10.0	1.5	4.20	1.34	16.21	0.39	0.16
Panel B: $\phi = 3.5$, $\rho = 0.979$						
7.5	0.5	1.11	4.80	14.17	1.17	0.10
7.5	1.5	3.29	1.61	18.23	0.39	0.19
10.0	0.5	2.07	4.89	14.17	1.17	0.10
10.0	1.5	5.10	1.34	18.23	0.39	0.19
Panel C: $\phi = 3.0$, $\rho = \psi_e = 0$						
7.5	0.5	-0.74	4.02	12.15	0.00	0.00
7.5	1.5	-0.74	1.93	12.15	0.00	0.00
10.0	0.5	-0.74	3.75	12.15	0.00	0.00
10.0	1.5	-0.74	1.78	12.15	0.00	0.00

5.2 Table II: asset-pricing implications, Case I

Read:

- When the IES is below one, the model struggles: for example, with $\gamma = 10$ and $\psi = 0.5$, the equity premium is only about 1.19% and the risk-free rate is too high at about 4.89%.
- When the IES is above one, the fit improves sharply: with $\gamma = 10$ and $\psi = 1.5$, the equity premium rises to about 4.20%, the risk-free rate falls to about 1.34%, market volatility rises to about 16.21%, and the price-dividend-ratio volatility rises to about 0.16.
- If dividends are made more sensitive to growth news (higher ϕ), both the equity premium and

valuation-ratio volatility rise further.

- In the i.i.d. case, the model produces essentially no valuation-ratio variation and even a negative equity premium.

Economic message. Persistent expected growth already goes a long way. But it only works quantitatively when the IES exceeds one, because only then do good long-run growth prospects raise asset prices rather than lower them.

5.3 Table III: properties of consumption volatility

Read:

- Variance ratios of realized consumption volatility rise with horizon: about 0.95 at two years, 1.26 at five years, and 1.75 at ten years.
- Current price-dividend ratios predict future realized volatility with negative coefficients: approximately -0.11 at a two-year horizon and around -0.10 at a five-year horizon.

Economic message. Volatility is persistent and valuations embed information about future uncertainty. This table is the empirical backbone of the stochastic-volatility channel.

Table III
Properties of Consumption Volatility

The entries in Panel A are the variance ratios ($VR(j)$) for $\{\epsilon_{g^a,t}\}$, which is the absolute value of the residual from the regression $g_t^a = \sum_{j=1}^5 A_j g_{t-j}^a + \epsilon_{g^a,t}$, where g_t^a denotes annual consumption growth rate. Panel B provides regression results for $\{\epsilon_{g^a,t+j}\} = \alpha + B(j)(p_t - d_t) + v_{t+j}$, and j indicates the forecast horizon in years. The statistics are based on annual observations from 1929 to 1998 of real nondurables and services consumption (BEA). The price-dividend ratio is based on the CRSP value-weighted return. Standard errors are Newey and West (1987) corrected using 10 lags.

Horizon	Panel A: Variance Ratios		Panel B: Predicting $\{\epsilon_{g^a,t+j}\}$		
	$VR(j)$	SE	$B(j)$	SE	R^2
2	0.95	(0.38)	-0.11	(0.04)	0.06
5	1.26	(1.09)	-0.10	(0.05)	0.04
10	1.75	(2.46)	-0.08	(0.08)	0.03

Table IV
Asset Pricing Implications—Case II

The entries are model population values of asset prices. The model incorporates fluctuating economic uncertainty (i.e., Case II) using the process in equation (8). In addition to the parameter values given in Panel A of Table II ($\delta = 0.998$, $\mu = \mu_d = 0.0015$, $\rho = 0.979$, $\sigma = 0.0078$, $\phi = 3$, $\phi_c = 0.044$, and $\phi_d = 4.5$), the parameters of the stochastic volatility process are $v_1 = 0.987$ and $\sigma_w = 0.23 \times 10^{-5}$. The predictable variation of realized volatility is 5.5%. The expressions $E(R_m - R_f)$ and $E(R_f)$ are, respectively, the annualized equity premium and mean risk-free rate. The expressions $\sigma(R_m)$, $\sigma(R_f)$, and $\sigma(p - d)$ are the annualized volatilities of the market return, risk-free rate, and the log price-dividend, respectively. The expressions $AC1$ and $AC2$ denote, respectively, the first and second autocorrelation. Standard errors are Newey and West (1987) corrected using 10 lags.

Variable	Data		Model	
	Estimate	SE	$\gamma = 7.5$	$\gamma = 10$
Returns				
$E(r_m - r_f)$	6.33	(2.15)	4.01	6.84
$E(r_f)$	0.86	(0.42)	1.44	0.93
$\sigma(r_m)$	19.42	(3.07)	17.81	18.65
$\sigma(r_f)$	0.97	(0.28)	0.44	0.57
Price Dividend				
$E(\exp(p - d))$	26.56	(2.53)	25.02	19.98
$\sigma(p - d)$	0.29	(0.04)	0.18	0.21
$AC1(p - d)$	0.81	(0.09)	0.80	0.82
$AC2(p - d)$	0.64	(0.15)	0.65	0.67

5.4 Table IV: asset-pricing implications, Case II

Read:

- In the data, the annual equity premium is about 6.33%, the risk-free rate about 0.86%, market volatility about 19.42%, and price-dividend-ratio volatility about 0.29.
- In the preferred model with $\gamma = 10$ and $\psi = 1.5$, the model delivers an equity premium of about 6.84%, a risk-free rate of about 0.93%, market volatility of about 18.65%, risk-free-rate volatility of about 0.57%, and price-dividend-ratio volatility of about 0.21.
- The model also reproduces the high persistence of the price-dividend ratio: $AC_1 \approx 0.82$ and $AC_2 \approx 0.67$.

Economic message. Adding stochastic volatility closes most of the remaining gap. This is the paper’s headline quantitative table.

5.5 Table V: decomposing the variance of the pricing kernel

Read:

- The annualized maximal Sharpe ratio in the model is about 0.73.
- Roughly 47% of pricing-kernel variance comes from expected-growth shocks.
- About 39% comes from stochastic-volatility shocks.
- Only about 14% comes from high-frequency i.i.d.-style consumption news.

Economic message. The model’s risk is genuinely long-run risk. Most of the priced variation comes from persistent growth and uncertainty shocks, not from transitory consumption noise.

Table V
Decomposing the Variance of the Pricing Kernel

Entries are the relative variance of different shocks to the variance of the pricing kernel. The entries are based on the model configuration described in Table IV with $\gamma = 10$. The volatility of the maximal Sharpe ratio is annualized in order to make it comparable to the Sharpe ratio on annualized returns.

Volatility of Pricing Kernel	Relative Variance of Shocks		
	Independent Consumption	Expected Growth Rate	Fluctuating Economic Uncertainty
0.73	14%	47%	39%

Table VI
**Predictability of Returns, Growth Rates,
and Price-Dividend Ratios**

This table provides evidence on predictability of future excess returns and growth rates by price-dividend ratios, and the predictability of price-dividend ratios by consumption volatility. The entries in Panel A correspond to regressing $r_{t+1}^e + r_{t+2}^e \cdots + r_{t+j}^e = \alpha(j) + B(j) \log(P_t/D_t) + v_{t+j}$, where r_{t+j}^e is the excess return, and j denotes the forecast horizon in years. The entries in Panel B correspond to regressing $g_{t+1}^c + g_{t+2}^c \cdots + g_{t+j}^c = \alpha(j) + B(j) \log(P_t/D_t) + v_{t+j}$, and g^c is annualized consumption growth. The entries in Panel C correspond to $\log(P_{t+j}/D_{t+j}) = \alpha(j) + B(j) |e_{g^c, j}| + v_{t+j}$, where $|e_{g^c, j}|$ is the volatility of consumption defined as the absolute value of the residual from regressing $g_t^c = \sum_{j=1}^5 A_j g_{t-j}^c + e_{g^c, j}$. The model is based on the process in equation (8), with parameter configuration given in Table IV and $\gamma = 10$. The entries for the model are based on 1,000 simulations each with 840 monthly observations that are time-aggregated to an annual frequency. Standard errors are Newey and West (1987) corrected using 10 lags.

Variable	Panel A: Excess Returns			Panel B: Growth Rates			Panel C: Volatility		
	Data	SE	Model	Data	SE	Model	Data	SE	Model
$B(1)$	-0.08	(0.07)	-0.18	0.04	(0.03)	0.06	-8.78	(3.58)	-3.74
$B(3)$	-0.37	(0.16)	-0.47	0.03	(0.05)	0.12	-8.32	(2.81)	-2.54
$B(5)$	-0.66	(0.21)	-0.66	0.02	(0.04)	0.15	-8.65	(2.67)	-1.56
$R^2(1)$	0.02	(0.04)	0.05	0.13	(0.09)	0.10	0.12	(0.05)	0.14
$R^2(3)$	0.19	(0.13)	0.10	0.02	(0.05)	0.12	0.11	(0.04)	0.08
$R^2(5)$	0.37	(0.15)	0.16	0.01	(0.02)	0.11	0.12	(0.04)	0.05

5.6 Table VI: predictability of returns, growth rates, and price-dividend ratios

Read:

- Panel A: higher dividend yields predict higher future excess returns. Because the regression uses the log price-dividend ratio, the coefficients are negative and become more negative at longer horizons. The model reproduces this pattern reasonably well.
- Panel B: price-dividend ratios have little ability to predict future consumption growth, both in the data and in the model.
- Panel C: realized consumption volatility predicts future price-dividend ratios negatively, with the data showing coefficients around -8.8 , -8.3 , and -8.7 at 1-, 3-, and 5-year horizons; the model gets the sign and broad pattern right, though the magnitudes are smaller.

Economic message. The model is strongest on return predictability and on the negative volatility-valuation link. It is intentionally much weaker on growth predictability, which is also weak in the data.

6 Short summary

Bansal and Yaron build an equilibrium asset-pricing model in which expected growth has a tiny but very persistent component and macroeconomic uncertainty is time varying. Those two ingredients, combined with Epstein–Zin preferences, make equity prices very sensitive to long-run growth news and volatility news. Quantitatively, the preferred version of the model comes close to matching the equity premium, the risk-free rate, the volatility and persistence of valuation ratios, and long-horizon return predictability. The paper’s central lesson is that asset prices are moved by *small but persistent* macro risks that are hard to detect in standard short-sample macro regressions but large enough to matter enormously for valuation.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a consumption-based model can explain many leading asset-pricing puzzles once it allows for two macroeconomic ingredients: persistent expected growth and stochastic volatility. The success of the model does not come from implausibly large consumption shocks. It comes from giving prices exposure to *long-run* movements in growth prospects and economic uncertainty.

7.2 Economic intuition

The intuition is simple but powerful. A one-time shock to expected growth is not just a one-year event if expected growth is persistent; it changes the entire future path of consumption and dividends. Likewise, a shock to uncertainty matters because it changes required compensation for risk over many future periods. Equities are especially sensitive because dividends are levered relative to consumption. If the IES is above one, investors want to own more wealth when long-run growth improves, so valuations rise. If uncertainty rises, discount rates rise and valuations fall.

7.3 Takeaway

This paper is one of the clearest demonstrations that “macro news” and “asset-pricing news” can be closely linked even when the macro variation looks tiny in standard time-series moments. The key is persistence. Small persistent risks can be almost invisible in short-run macro data but still dominate the valuation of long-lived assets. That is why the long-run-risks framework became so influential after this paper.